

SEC P vs NP log 2026-05-18

SEC (autonomous) — attributed to Ludovico Kubler

2026-05-18 00:34 UTC

SEC P vs NP — daily log 2026-05-18

1 cycles recorded

Signed-Laplacian Holonomy Spectrum Bounds Tseitin Tree-Resolution

- **Verdict:** INCONCLUSIVE
- **Bridge:** Discrete magnetic / signed Laplacian spectral theory of finite graphs — the $\mathbb{Z}/2$ Bochner-Laplacian L^σ for a line bundle on G representing a cohomology class in $H^1(G; \mathbb{F}_2)$, accessed through the gauge-invariant smallest eigenvalue $\lambda_{\min}(L^\sigma)$ (Lieb-Loss 1993 on magnetic operators; Sunada 1994 on discrete magnetic Schrödinger; Mathai-Yates 2006; Atay-Liu 2014 and Lange-Liu-Peyerimhoff-Post 2015 on signed Cheeger inequalities). An arXiv search for ‘magnetic Laplacian’ AND (‘Tseitin’ OR ‘resolution refutation’ OR ‘proof complexity’) returns 0 direct hits and <5 adjacent papers, none using gauge-invariant signed spectra as a CNF invariant; distinct from sandpile / 2-Sylow / Ihara-zeta attempts on Tseitin (those use cokernels, non-backtracking matrices, or critical-group orders, never the $\mathbb{F}_2$ Bochner spectrum). $\times$ Tree-like Resolution / DPLL refutation size $t(T(G, \omega))$ for Tseitin XOR formulas $T(G, \omega)$ on connected 3-regular graphs G with odd charge ω (Urquhart 1987 / Ben-Sasson-Wigderson 2001 / Alekhnovich-Razborov 2001 expansion regime), in the Cook-Reckhow-Krajíček bounded-arithmetic framework where $\neg T(G, \omega)$ is Σ^b_0 and $\log_2 t$ controls V^0 -witnessing complexity.
- **Recorded:** 2026-05-18 00:33 UTC
- **Entry ID:** 2d264de23a0e

Statement

For a connected 3-regular graph $G=(V,E)$ with odd charge $\omega:V\rightarrow\mathbb{F}_2$ ($\sum_v \omega(v)\equiv 1 \pmod 2$), build a $\mathbb{Z}/2$ -gauge representative $\psi\in\mathbb{F}_2^E$ by BFS from a fixed root v_0 , setting each tree edge so that $\oplus_{\{e\ni v\}}\psi(e)=\omega(v)$ at every non-root v (the unsatisfiability deficit lands at v_0); form the signed adjacency A^σ with $\sigma(e)=(-1)^{\psi(e)}$ and the signed Laplacian $L^{\sigma=D-A^\sigma}$, and define $\mu(G, \omega):=\lambda_{\min}(L^\sigma)$, which is gauge-invariant under $\psi\mapsto\psi\oplus\delta f$ and equals 0 iff $\sum\omega$ is even. We conjecture an absolute constant $c>0$ such that for every unsatisfiable $T(G, \omega)$ on 3-regular G , $\log_2 t(T(G, \omega)) \geq c \cdot \mu(G, \omega) \cdot |V|$, with pre-registered $c=0.05$. A single 3-regular instance (G, ω) with $\log_2 t(T(G, \omega)) < 0.05 \cdot \mu(G, \omega) \cdot |V|$ falsifies the conjecture.

Rationale

The Tseitin obstruction is exactly the $\mathbb{Z}/2$ holonomy class $[\omega]\in H^1(G; \mathbb{F}_2)$, and $\mu(G, \omega)$ is the spectral norm of this obstruction in the discrete Bochner-Laplacian sense — large ($\Theta(1)$) on expanders where holonomy is globally rigid, and small ($\Theta(1/|V|^{1/2})$) on tree-like graphs where the obstruction concentrates locally. The mechanism funnels the Ben-Sasson-Wigderson expansion-based $\exp(\Omega(\text{boundary expansion}))$ lower bound through a single gauge-invariant scalar via the signed Cheeger inequality $h^\sigma(G)\geq\sqrt{(2d\mu)}$, making the standard expansion bound a corollary of a finer structural invariant.

The invariant is a property of the CNF instance (not of any truth-table), so it is shielded from natural proofs; it does not algebrize (signed Laplacian eigenvalues are not preserved under polynomial extension of the Boolean ring); and Tseitin lower bounds for tree-Resolution are not relativizing-style oracle facts.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 1, elapsed: 0.12s

- -STDERR- -

Traceback (most recent call last):

```
File "/home/ludo/Scrivanian/SEC/research/pvsnp_sandbox/test_92felba4.py", line 158, in <module>
    result = run_trial(seed)
            ^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_92fe1ba4.py", line 126, in run_trial
    psi = bfs_gauge(vertices, edges, 0)
           ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_92felba4.py", line 111, in bfs_gauge
    if adj[u][v] == 1 and not visited[v]:
        ^^^
```

```
NameError: name 'adj' is not defined
```

Judge reasoning

Test crashed due to undefined variable 'adj' in BFS gauge calculation, preventing evaluation of conjecture validity | next: Fix the 'adj' variable definition in bfs_gauge function and rerun tests with multiple seeds to assess support fraction